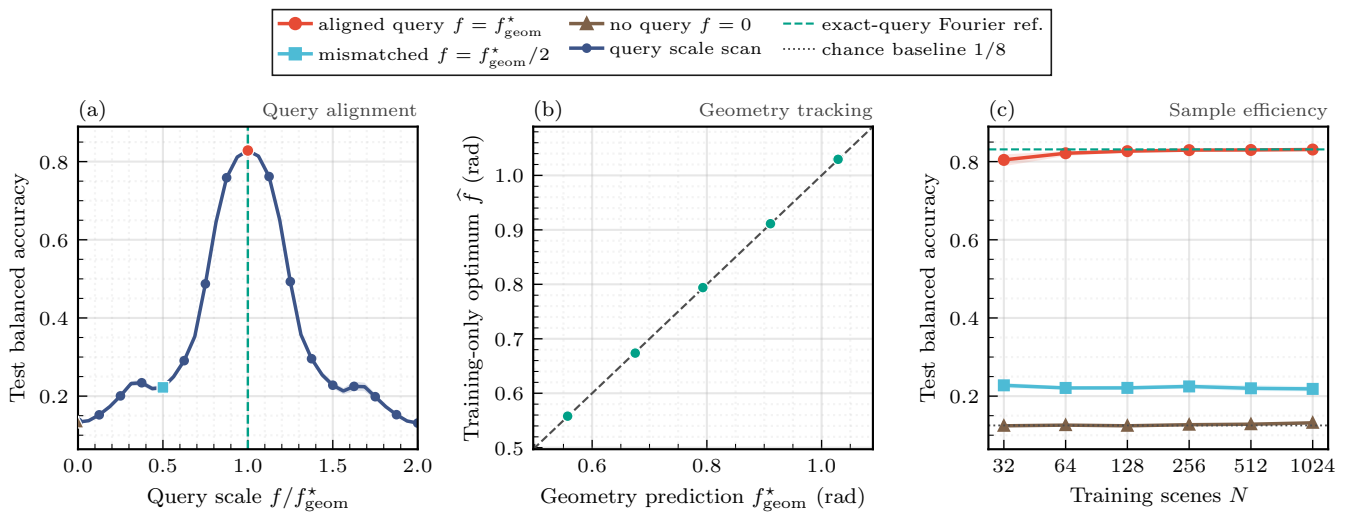

aligned query $f = f^\star_{\mathrm{geom}}$
mismatched $f = f^\star_{\mathrm{geom}}/2$
no query $f = 0$
query scale scan
exact-query Fourier ref.
chance baseline 1/8
(a)
Query alignment
Test balanced accuracy
Query scale $f/f^\star_{\mathrm{geom}}$
(b)
Geometry tracking
Training-only optimum $\hat{f}$ (rad)
Geometry prediction $f^\star_{\mathrm{geom}}$ (rad)
(c)
Sample efficiency
Test balanced accuracy
Training scenes N